

Dataset Summary

Metric	Value
Total features (columns)	20
Total rows (transactions)	18,763
Unique customers	4,754
Missing values	Only in purchase_amount (657) and review_rating (563) – imputed by category median
Data types	3 integers, 2 floats, 15 strings (categorical)
Time coverage	No exact dates; seasons: Fall, Spring, Summer, Winter

Feature Definitions

customer_id – Unique integer identifier for each customer. Range: 1 to 4,754.

age – Customer age in years. Integer. Range: 18 to 70.

gender – Categorical: 'Female' or 'Male'.

item_purchased – Product name. One of 25 items (e.g., Boots, Jewelry, Handbag, Shirt, Dress, Coat, etc.).

category – Product category. One of: 'Accessories', 'Clothing', 'Footwear', 'Outerwear'.

purchase_amount – Transaction value in USD. Float. Range: 12.00 to 384.00. Missing values imputed by category median.

location – U.S. state. All 50 states appear.

size – Product size. One of: 'S', 'M', 'L', 'XL'.

color – Product color. One of 25 colors (e.g., Black, Blue, Brown, Lavender, Red, White).

season – Season of purchase. One of: 'Fall', 'Spring', 'Summer', 'Winter'.

review_rating – Customer rating (2.7 to 5.0). Float. Missing values imputed by category median.

subscription_status – 'Yes' or 'No'.

shipping_type – One of: 'Standard', 'Free Shipping', 'Express', '2-Day Shipping', 'Store Pickup', 'Next Day Air'.

discount_applied – 'Yes' or 'No'.

promo_code_used – 'Yes' or 'No'.

previous_purchases – Historical purchase count. Integer. Range: 0 to 85.

payment_method – One of: 'Credit Card', 'Debit Card', 'PayPal', 'Venmo', 'Bank Transfer', 'Cash'.

frequency_of_purchases – Self-reported frequency. One of: 'Weekly', 'Bi-Weekly', 'Fortnightly', 'Monthly', 'Quarterly', 'Every 3 Months', 'Annually'.

age_group (engineered) – Binned age: '18-30', '31-45', '46-60', '61+'.

purchase_frequency_days (engineered) – Numeric days: Weekly=7, Bi-Weekly/Fortnightly=14, Monthly=30, Quarterly/Every 3 Months=90, Annually=365.